



JERUSALEM COLLEGE OF ENGINEERING, CHENNAI – 600 100
(An Autonomous Institution Affiliated to Anna University, Chennai)

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Question Paper Code : 225024

B.E./B.Tech. END SEMESTER EXAMINATIONS, APRIL/MAY 2025

Fourth Semester

Computer Science and Engineering

JMA2402 – APPLIED LINEAR ALGEBRA

(Common to Computer Science and Engineering (Artificial Intelligence and Machine Learning), Computer Science and Engineering (Cyber Security), Electronics and Communication Engineering, Artificial Intelligence and Data Science, Computer Science and Business Systems, and Information Technology)

(Regulation 2023)

Time : Three hours

Maximum : 100 marks

Answer ALL questions

PART A – (10 × 2 = 20 marks)

1. Define rank of a matrix.
2. What is the condition for a system of linear equations to have (i) exactly one solution (ii) infinitely many solutions.
3. Define vector space.
4. Is $V = \{(a_1, a_2, a_3) \in \mathbb{R}^3 / a_1 + 2a_2 - 3a_3 = 1\}$ a subspace of vector space $\mathbb{R}^3$ defined over the field of real numbers $\mathbb{R}$ under coordinate wise addition and scalar multiplication?
5. Define linear transformation.
6. Is $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1 + a_2, a_1)$ one-one?
7. If T is a linear transformation of $\mathbb{R}^2$ defined by $T(a, b) = (a, 0)$, write the matrix of T relative to the standard basis of $\mathbb{R}^2$

8. Write the test for diagonalizability.
9. Define inner product space.
10. Show that orthogonality of vectors is a symmetric relation.

PART B – ($5 \times 16 = 80$ marks)

11. (a) (i) Test the consistency of the system of equations and solve if consistent.

$$x_1 + 2x_2 - x_3 - 5x_4 = 4$$

$$x_1 + 3x_2 - 2x_3 - 7x_4 = 5$$

$$2x_1 - x_2 + 3x_3 = 3$$

- (ii) Find the values of k for which the equations

$$x + y + z = 1, \quad x + 2y + 4z = k, \quad x + 4y + 10z = k^2$$

may have a solution. For these values of k find the solution also.

Or

- (b) (i) Solve the following system of equations by Gauss Jordan method.

$$x + y + z = 9$$

$$2x - 3y + 4z = 13$$

$$3x + 4y + 5z = 40$$

- (ii) Find the values of k for which the equations

$$kx + y + z = 1, x + ky + z = 1, x + y + kz = 1$$

have (a) infinitely many solutions (b) no solution (c) unique solution

12. (a) Let $V = \{(a_1, a_2, \dots, a_n) / a_i \in \mathbb{R}, i = 1, 2, \dots, n\}$. Is V a vector space over $\mathbb{R}$ with the operations coordinate wise addition and scalar multiplication?

Or

- (b) (i) Check whether the set of vectors are linearly dependent, if so express any one vector as a linear combination of other vectors $\{(-2, 2, 2), (1, 2, -1), (-3, -3, 3)\}$

- (ii) Determine whether the given vector v_1 is in the span of S

$$v_1 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix} \quad S = \left\{ \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\}$$

13. (a) (i) Define reflection about x -axis and projection on x -axis and show that they are linear transformations.

- (ii) If $T: V \rightarrow W$ is a linear transformation, where V and W are vector spaces, then prove that T is one-one if and only if $N(T) = \{0\}$.

Or

- (b) Prove that $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ is a linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify dimension theorem. Also check if T is one-one or onto

14. (a) (i) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation defined by

$$T(a_1, a_2) = (2a_1 - a_2, 3a_1 + 4a_2, a_1),$$

find $[T]_{\alpha}^{\beta}$ given that α & β are the standard basis of $\mathbb{R}^2$ & $\mathbb{R}^3$ respectively.

- (ii) Find the matrix of the linear transformation T on $\mathbb{R}^3$ defined by $T(a, b, c) = (2b + c, a - 4b, 3a)$ with respect to the ordered basis $\beta = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$

Or

- (b) If T is linear on $V = \mathbb{R}^3$ defined by

$$T(a, b, c) = (7a - 4b + 10c, 4a - 3b + 8c, -2a + b - 2c),$$

find the eigen values of T and an ordered basis β for V such that $[T]_{\beta}$ is a diagonal matrix.

15. (a) (i) Show that $F^n(\mathbb{C})$ is an inner product space with respect to the inner product defined as $\langle x, y \rangle = \sum_{i=1}^n x_i \bar{y}_i$ where $x = (x_1, x_2, \dots, x_n)$ & $y = (y_1, y_2, \dots, y_n)$

- (ii) State and prove Cauchy-Schwartz inequality.

Or

- (b) Apply Gram-Schmidt orthogonalization process to the subset $S = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ of $\mathbb{R}^3$ to obtain an orthonormal basis. Also find Fourier coefficients of $(1, 0, 1)$ relative to this orthonormal basis.